



Mechatronic Sensors Trainer

User Manual – Setup and Configuration

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This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

FCC Notice This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

Industry Canada Notice This Class A digital apparatus complies with CAN ICES-3 (A). Cet appareil numérique de la classe A est conforme à la norme NMB-3 (A) du Canada.



Waste Electrical and Electronic Equipment (WEEE)

This symbol indicates that waste products must be disposed of separately from municipal household waste, according to Directive 2012/19/EU of the European Parliament and the Council on waste electrical and electronic equipment (WEEE). All products at the end of their life cycle must be sent to a WEEE collection and recycling center. Proper WEEE disposal reduces the environmental impact and the risk to human health due to potentially hazardous substances used in such equipment. Your cooperation in proper WEEE disposal will contribute to the effective usage of natural resources.

This product meets the essential requirements of applicable European Directives as follows:

CE Compliance 

- 2014/30/EU; Electromagnetic Compatibility Directive (EMC)
- 2014/30/EU; Electromagnetic Compatibility Directive (EMC)
- 2014/53/EU; Radio Equipment Directive (RED)

Warning: This is a Class A product. In a domestic environment this product may cause radio interference, in which case the user may be required to take adequate measures.



ESD Warning. The QBot Platform internal components are sensitive to electrostatic discharge. Before handling the QBot Platform, ensure that you have been properly grounded.

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A. Introduction

The Quanser Mechatronic Sensors Trainer is an integrated sensor experiment. It is designed to help teach fundamental sensor concepts and theories on an easy-to-use and intuitive platform.

The compact device includes a rich suite of distance, motion, environmental, light, force, and touch sensors, allowing students to not only learn the fundamental workings of these sensors, but also develop key skills in component selection and making design decisions for complex measurement and perception systems.

With the ability to provide real-time measurements, the Mechatronic Sensors Trainer can be integrated into larger systems or used to enhance the functioning capabilities of other Quanser systems. In addition, this device can also be used to measure and store data asynchronously for offline processing at a later time.

The unit is intended for lab and classroom use to support fundamentals labs, guided integration challenges, and open-ended projects.



Figure 1. Mechatronic Sensors Trainer

B. Hardware Components

The main components of the Sensors Trainer are listed in Table 1 and 2. These components are marked in Figures 2 and 3, which show the front, and side views of the Trainer.

The datasheet for each of the components is located in the same folder as this user manual.

i. Parts of the Device



Figure 2. Front of Mechatronic Sensors Trainer

ID	Component	ID	Component
1	Passive Infrared Sensor and Cap	6	RGB and IR LEDs
2	Radar Sensor	7	Color Ambient Sensor
3	Weather Sensor	8	Camera
4	Ultrasonic Sensor with Cone	9	Light Dependent Resistor
5	Time of Flight Sensor	10	IMU Sensor
		11	Infrared Distance Sensor

Table 1. Components in the front of the device

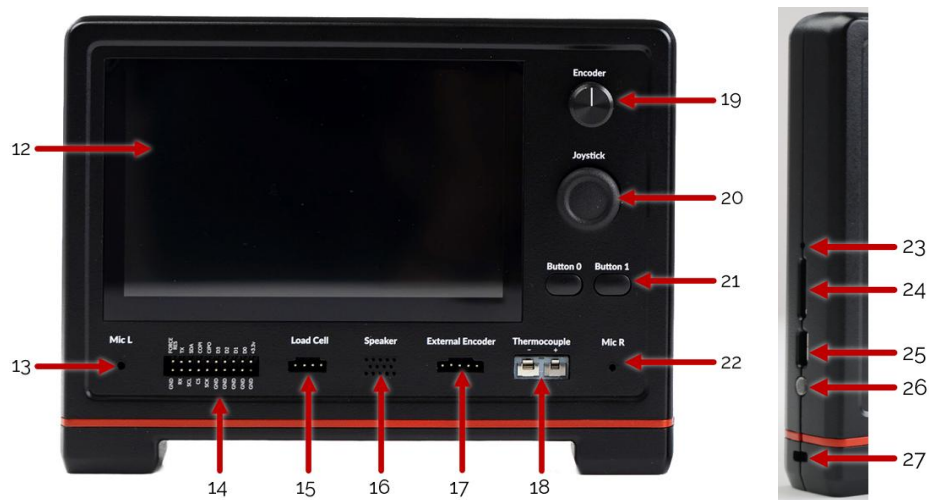


Figure 3. Back and Side of Mechatronic Sensors Trainer

ID	Component	ID	Component
12	LCD Touch Sensor	20	Joystick
13	Left Microphone	21	Push Buttons (0 and 1)
14	GPIO	22	Right Microphone
15	Load Cell Connector	23	Programming Pin
16	Speaker	24	SD Card Slot
17	External Encoder/PWM Connector	25	USB C Connector
18	K-Type Thermocouple Connector	26	Status LED
19	Encoder Knob	27	Standard (K type) Kensington Lock

Table 2. Components in the front and side of the device

ii. External Provided Parts

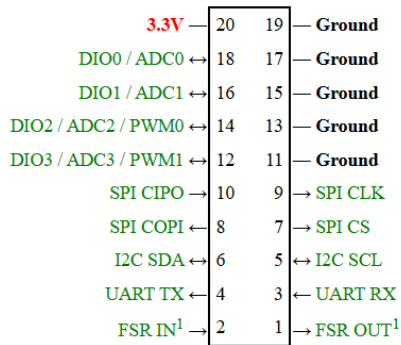
Pic of the 3 external sensors and how they get connected

iii. Connectors and GPIO

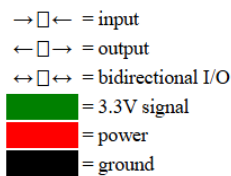
The Quanser Mechatronic Sensors Trainer has a number of connectors for sensors or expansion I/O. These connectors and their pinouts are listed below.

Note that all pins are 3.3V and are *not* 5V tolerant unless marked otherwise. The analog inputs only support 0 to +3.3V. *Exceeding these voltages may damage the board!*

User I/O Connector (GPIO)



Legend

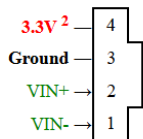


Thermocouple Connector



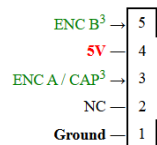
¹ Dedicated Force Sensing Resistor (FSR) pins

Load Cell Connector



² Dedicated Load Cell power supply. Do not use for anything else.

Encoder Connector



³ 5V tolerant input pins

iv. Dimensions

Dimension	Size
Length	18 cm
Width	3 cm
Height	13 cm

v. Capabilities

The Quanser Mechatronic Sensors Trainer provides the following capabilities:

- Analog joystick with X and Y axes (16-bit resolution) and button
- Analog inputs (four optional single-ended channels 0.0V to 3.3V, 16-bit resolution)
- Camera (RGB, up to 2592x1944 resolution, OV5640 sensor)
- Color and ambient light sensor (including clear, R, G, B and infrared, VEML3328)
- CPU temperature
- Current sense on external +3.3V supply (INA226)
- Digital I/O (four channels)
- External encoder (with 1X, 2X and 4X quadrature)
- Force sensing resistor (16-bit resolution)
- IMU with 3-DOF gyroscope (up to 2000 °/s), 3-DOF accelerometer (up to ±16g), 3-DOF magnetometer (up to 5000 µT) and temperature (ICM-20948)
- Infrared distance sensor (16-bit resolution)
- Input capture measuring period and pulse width (one optional channel)
- Knob encoder (with 1X, 2X and 4X quadrature and A/B signals)
- LCD display with touch panel (800x480 resolution, 120 Hz touch panel up to 10 fingers)
- LEDs to provide illumination for the color sensor (two RGB and infrared LEDs)
- Light-dependent resistor (16-bit resolution)
- Load cell sensor (24-bit resolution, NAU7802)
- Passive infrared sensor (16-bit resolution, PL-Q873-02)
- Pushbuttons (two buttons with configurable polarity)
- PWM outputs (two optional channels up to 16-bit resolution and up to a 120 MHz PWM)
- Radar (includes distance, presence, power bins, envelope, IQ and sparse measurements, with up to 14455 points, A111-001-T&R)
- Speaker (96 kHz audio, MAX98357AETE+T)
- Stereo microphone (48 kHz 24-bit audio, ICS-41351)
- Thermocouple (internal & external temp, fault detection, MAX31855KASA+T)
- Time of flight sensor with up to 64 zones (includes distance, sigma, reflectance and targets detected per zone, VL53L8CH)
- UART for external serial devices (up to 15 Mbaud)
- Ultrasonic sensor (includes distance and up to 450 IQ data points, CH201-00ABR)
- Weather sensor (pressure, temperature with up to 20-bit resolution and humidity with 16-bit resolution, BME280)
- Game controller overrides for all point-of-view, buttons, joysticks and bumpers

vi. Mechatronic Sensors Trainer as Devices in a PC

When the Quanser Mechatronic Sensors Trainer is connected to a PC it will show up as multiple devices. In particular, it will appear in Device Manager in Windows as the following devices (or possible variations thereof):

- Audio inputs and outputs/Microphone (2-Quanser Mechatronic Sensors Trainer)
- Audio inputs and outputs/Speakers (Quanser Mechatronic Sensors Trainer)
- Cameras/Quanser Mechatronic Sensors Trainer
- Ports (COM & LPT)/USB Serial Device (COM nn)
- Universal Serial Bus devices/Quanser Mechatronic Sensors Trainer HIL
- Universal Serial Bus devices/Quanser Mechatronic Sensors Trainer I2C
- Universal Serial Bus devices/Quanser Mechatronic Sensors Trainer LCD
- Universal Serial Bus devices/Quanser Mechatronic Sensors Trainer SPI
- Xbox 360 Peripherals/Xbox 360 Controller for Windows

The Quanser Mechatronic Sensors Trainer conforms to the specifications for USB audio devices, a USB camera and a USB serial port so it may be used with operating systems and software that supports standard USB devices when it comes to those devices (microphones, speaker, camera and serial port). For example, the Camera application that comes with Windows supports Quanser Mechatronic Sensors Trainer out of the box, while PuTTY supports the serial port on Quanser Mechatronic Sensors Trainer out of the box. Likewise, the "Setup USB game controllers" Properties dialog supports Quanser Mechatronic Sensors Trainer as a game controller out of the box.

vii. SD Card Considerations

To use an SD Card with the Mechatronic Sensors Trainer, the SD card must be formatted with the FAT file system (FAT32) as opposed to exFAT. 32 GB or less tends to be FAT filesystem by default, anything larger is exFAT by default. However, we recommend formatting the SD card to the desired file system to be sure. To read an SD card larger than 32 GB that has been formatted as FAT, an updated Windows 11 computer is needed.

We recommend using a 32 GB SD card or smaller for this device. You can find information on how to format an SD card using a Windows computer [here](#).

viii. Encoder Standards

Encoder outputs are assigned to match the standard across Quanser devices, which could lead to a mismatch with the datasheet. The standard in Quanser devices is B leads A for a CW direction.

When motors are used in the device, a positive motor input leads to a counterclockwise rotation of the shaft which increases the encoder counts. This matches all previously released Quanser legacy devices.

ix. Camera Resolutions

The Quanser Mechatronic Sensors Trainer provides a color camera (OV5640) with up to 2592x1944 pixel resolution. The available camera resolutions are listed below:

Resolution	Description
160x120	Smallest resolution, 0.02MP 4:3 aspect ratio.
176x144	0.03MP 11:9 aspect ratio.
320x240	0.08MP 4:3 aspect ratio.
640x480	0.3MP 4:3 aspect ratio.
720x480	0.3MP 3:2 aspect ratio.
720x576	0.4MP 5:4 aspect ratio.
1024x768	0.8MP 4:3 aspect ratio.
1280x720	0.9MP 16:9 aspect ratio.
1920x1080	2.1MP 16:9 aspect ratio.
2592x1944	Maximum resolution of 5.0MP 4:3 aspect ratio.

C. Handling and Setup

i. Powering Up the Device

To set up the Mechatronic Sensors Trainer, follow these steps:

1. Make sure the Mechatronic Sensors Trainer is out of its case. Add the base underneath if needed, make sure the notch on the base matches the notch on the sensor side of the device.
2. Power the device up using the provided USB cable connected to the PC.
3. The device should start up with 3 blinks in blue and change colors quickly to a constant white.
4. The LED next to the USB C port will stay white after the display shows that it is loading.
5. The load screen with the Quanser Logo should disappear after a few seconds and you should see some evenly spaced crosses on a black background on the touch screen.

ii. Powering Up the Device Without a Computer

When the device is powered by a standalone power source, such as a USB power adapter or battery pack, and not connected to a computer (for example, a Windows PC or Raspberry Pi), the power-up procedure is the same as described in the previous section.

In this configuration where there is no computer detected, the device automatically enters **standalone recording mode**, which allows data from all sensors to be recorded directly to an SD card. On the bottom-right corner of the screen, a **REC** button will be displayed. If no

compatible SD card is detected, the **REC** button appears greyed out. If an SD card is inserted and formatted according to the SD Card Considerations section (B.vi), the **REC** button becomes active, and the text will be white. Press **REC** to begin recording all signals to the SD card. Once recording has started, the button label changes to **STOP**. Press **STOP** to end the recording.

You may create multiple recordings during a session. Each recording is saved with a unique filename to prevent existing data from being overwritten.

iii. Powering Off the Device

The Mechatronic Sensors Trainer does not have a power off button. To power off the device, simply disconnect the USB cable the device is connected to.

iv. Low Power Mode

When using the device with the provided USB A to USB C cable, the screen will display a Low-Power mode warning on the display screen. To avoid this, use the provided USB C to USB C cable instead.

In low power mode, the maximum current that may be drawn from the user 3.3V and encoder 5V is 0.1 A total. In normal mode, it is 0.5 A. This only affects external sensors, all the onboard sensors are not affected when this warning is shown on the screen.

If the current limit is exceeded, you will get a fatal error message on the LCD requesting that you power down the device and resolve the issue.

D. Environmental

The Mechatronic Sensors Trainer is designed to function under the following environmental conditions:

- Standard rating
- Indoor use only
- Atmospheric conditions
 - Temperature - 15°C to 35°C
 - Altitude - up to 2000 m
 - Relative humidity - 30% to 60%
 - Air Pressure – 86 kPa (860 mbar) – 106 kPa (1060 mbar)
- Pollution Degree 2
- Mains supply voltage fluctuations up to 10% of nominal voltage
- Maximum transient overvoltage 2500 V

Commented [LG1]: Please verify this and change if needed. I assume that temperature is wrong and the rest is too

E. Electrical Considerations



ESD warning

The QArm end-effector components are sensitive to electrostatic discharge. Before handling the board, ensure that you have been properly grounded.



Caution

Always monitor the current draw and motor temperature to ensure that you do not exceed the maximum current draw or 50°C.



Caution

Do not have conductive material touch the end-effector IO board as it can short and damage the electronics.



Caution

The QArm is not waterproof.

Commented [LG2]: Please verify this. These we copied from qarm. Feel free to just comment here what needs to be considered and I can format it as needed

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